Electric Vehicle Dashboard

Contents

1. Introduction

The purpose of this report is to present the findings of a Market Intelligence Dashboard developed to evaluate demand trends, the evolving global competitive landscape, and the effects of recent oil shocks on the electric vehicle (EV) industry. The report first examines increasing competitive pressure within the EV market, before analysing changes in consumer demand associated with oil shock events. Finally, it evaluates the emergence of Chinese EV manufacturers and their value proposition, with particular consideration of whether their competitive strategy prioritises production volume over product quality.

2. Rising Competitive Pressures

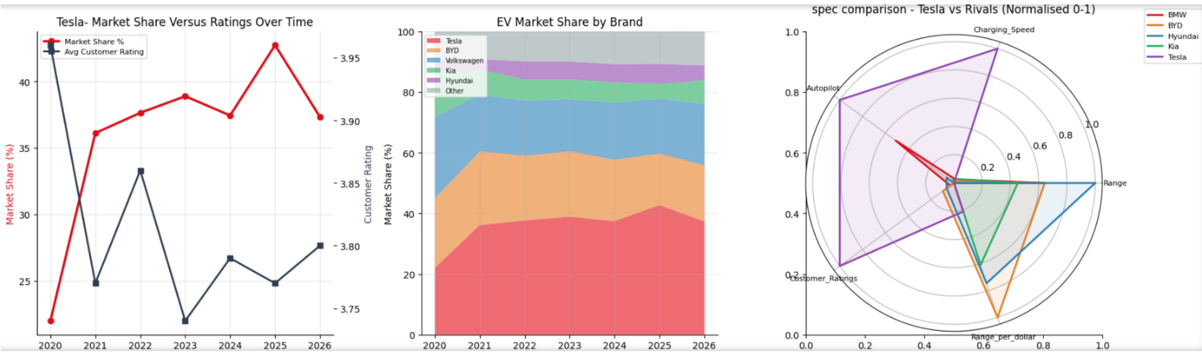


Chart A, Chart B, Chart C (From left to right)

The first section of this dashboard considers rising competitive pressures in the EV industry and is broken down into three questions in order to provide a comprehensive analysis on the industry.

In our first dashboard we will look at:

- As Tesla has expanded their offerings to serve multiple segments, has its average customer ratings held, or has quality perception diluted as volume and variety scaled significantly over recent years? (2.1)
- Has Tesla's share of Total Annual Sales declined Year-over-Year since 2022, if so, at what rate? (2.2)
- Have competitors closed the gap on metrics that Tesla has previously led and built its global reputation on, such as range, charging speed, and autopilot ability. (2.3)

2.1 Tesla – Market share versus Ratings

Beginning in 2020, Tesla saw its highest Average Customer Ratings while possessing their lowest market share at sub-25%, entering into 2021 – Tesla captured a large portion of the EV industry, likely benefiting from the 2021 Russo-Ukrainian Invasion and Oil shocks. During this initial surge in market share, customer ratings dropped in a near inversely-proportional manner – spiking upwards slightly in 2022 while Market Share moved upwards modestly.

In the Post-2022 region, the relationship between Market Share and Ratings has stabilised but still visibly reacts in opposite directions from eachother – potentially implying that Tesla's competitive gains are achieved through factors other than customer satisfaction or that Tesla's improvements in product quality and customer perception no longer translates into market share gain. Possibly being a consequence of lower-rated manufacturers entering the EV industry with aggressive pricing strategies and large-volume production, rather than being reliant on customer satisfaction as a means of expansion.

2.2 Tesla's YoY Market share

Tesla's market share has not declined since 2022 – it has grown. Share moved from 22.0% (2020) to 37.6–38.9% through 2022–2023, dipped slightly to 37.4% in 2024, then reached its high point of the dataset at 42.7% in 2025. The only year showing a real drop is 2026, down to 37.3% – but 2026 is a partial year in this dataset (roughly a third of the row count of 2025), so that dip is most likely a data-completeness artifact rather than a genuine reversal, and shouldn't be read as the start of a decline until a full year of 2026 data is available.

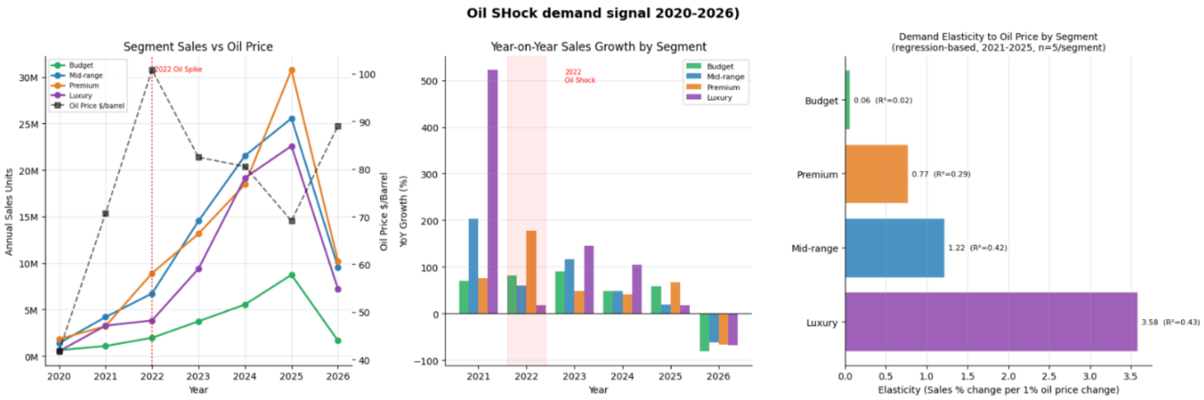
The competitive pressure in this chart shows up elsewhere: the combined share held by brands outside the top 6 (Tesla, BYD, Volkswagen, Kia, Hyundai, BMW) has crept up gradually across the whole period – from 6.7% in 2020 to 8.8% by 2026 – rather than surging in any single year. And within the top 6 itself, the pressure is uneven: Kia, an established manufacturer present in the data since 2020, has actually lost significant ground, falling from 14.7% share in 2020 to under 8% in most recent years. So this chart doesn't show an impending consolidation, nor a flagship rival closing in on Tesla specifically – it shows Tesla holding and extending its lead, while competitive pressure builds slowly at the margins through a lengthening tail of smaller entrants and a reshuffling among the established players themselves.

2.3 Competitors Specs

Looking at the Radar Chart (includes 5 largest companies) above, it is apparent that no manufacturer has begun to meet or lead Tesla on specifications such as Charging Speed, Autopilot, and their Customer Ratings.

However, Tesla falls short on Range, and Range_per_dollar when compared to the other four of the five largest companies in the EV industry. Possibly showing that factors such as range and range value is not a prioritised factor for Tesla when it comes to their vehicles and their vehicle innovation, as much as more “technical” features are, such as Autopilot and Charging Speed.

3. Demand Changes due to Oil Shocks



3.1 Segment Sales vs Oil Price

Every segment (Budget, Mid-range, Premium, and Luxury) grew every year from 2020-2025, regardless whether oil prices were rising (2021-2022) or falling (2023-2025). The overall structural growth curve of the EV market is the dominant signal in the chart above; oil prices move independently of it rather than being a visible factor that steers the trajectory. However, in 2022, Premium segment growth spiked significantly by over 170% - the same year that oil peaked – showing consistency with strong demand for higher-end EVs during years aligned with oil spiked even if the multi-year pattern does not hold up as a clean overall relationship (3.3).

3.2 YoY Sales Growth by Segment

Growth decelerated in the two most recent years complete years (2024, 2025) across segments, compared with the 2021 to 2023 window, which showed several segments posting triple digit growth.

- Luxury is the most extreme figure in this dataset at 522%, however, this is following small 2020 base sales of 525,769 units sold– rather than a demand signal, it may be more appropriate to treat this figure as a base-effect from small previous year sales.

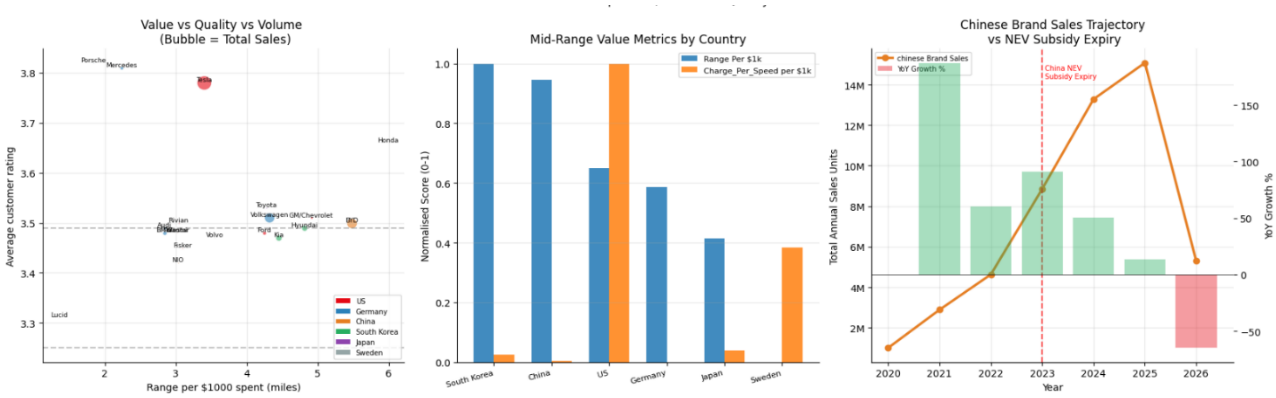
3.3 Demand Elasticity to Oil Price by Market Segment

The dashboards original elasticity calculation (Luxury -7.98, Budget -4.91, Mid-range -4.33, Premium -3.59) using a means-of-ratio approach suggested sales fell across every segment as the price of oil rose, the opposite of the hypothesis stated earlier that fuel savings was a driving factor to EV adoption.

Using a regression-based calculation, and excluding 2026's data from the elasticity calculation – we begin to see a chart that is more directionally consistent with our hypothesis (Luxury +3.58, Mid-range +1.22, Premium +0.77, Budget +0.06), showing that sales grew faster across segments as oil prices rose.

This result should still be read cautiously. With only five complete years of data per segment, R^2 ranges from just 0.02 (Budget) to 0.44 (Luxury), and none of the segments reach conventional statistical significance (p-values between 0.23 and 0.81). The corrected calculation is a meaningfully better estimate than the original - but the sample is too small to confirm a real oil-price effect either way. The fair conclusion is that this dataset shows no evidence of the negative relationship the original calculation implied, not that a positive relationship is confirmed.

4. Chinas Value Proposition



4.1 Value Vs. Quality Vs. Volume

BYD offers a strong value proposition, 5.48 range on miles/$1,000 second to Honda at 5.99, although BYD out scales Honda in volume significantly 50.5M units to 17,633 units, trailing only Tesla in volume. BYD's average price sits in the middle at $60,085 – cheaper than Tesla which averages at $93,523 and competing with the likes of Kia ($65,836) and Hyundai ($60,455). Its customer rating (3.50) is near the datasets overall median of 3.57, this points to BYD's effective edge coming from range-per-dollar rather than being the cheapest or highest-rated model on the market.

4.2 Mid-Range Value by Country

Within the mid-range segment, South Korea (5.44 range-miles per $1,000) and China (5.38) are essentially tied for the best value, both clearly ahead of the U.S. (5.05), Germany (4.98), and Japan (4.79) – and well ahead of Sweden, which trails on every metric in this segment: the lowest range-per-dollar (4.34), the lowest average range, and the lowest customer rating of the six countries compared. This directly answers the "China vs West" question for mid-range tier vehicles: yes, Chinese (and South Korean) manufacturers of EVs offer a measurably superior range-per-dollar value than every Western counterpart in the mid-range segment.

4.3 Chinese Brand Sales Trajectory and Subsidy Expiration

Chinese brand (BYD, Nio, Xiaomi) sales growth decelerated steadily from 2021 to 2025. Although, in 2023 growth accelerated – the year of the NEV subsidy expiry, rather than dropping immediately. Real levels of deceleration show up after a one-to-two-year lag, appearing in 2024-2025. This shows that there is no direct or sharp impact on sales growth in 2023 as a result of the NEV subsidy expiring; rather it more clearly displays the normalising growth rates in the maturing Chinese EV market.

5. Conclusion

Across all three questions, the dataset pushes back on some of the report's original hypotheses in useful ways: Tesla's market share grew rather than eroded over 2020-2025, even as its ratings decoupled from market share gains; the oil-shock elasticity signal only turns directionally positive once a calculation error is corrected, and even then isn't statistically conclusive on five years of data; and China's subsidy expiry doesn't produce an immediate dive in sales growth, just a gradually maturing growth rate. Where the original hypotheses do hold up well are: Tesla's spec profile and BYD's value proposition – both confirmed by the charts and numbers.